Lctr 6: Discrete-Time LTI Systems and Convolution Sum

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Focus: Section 2.1 (Pages 76 to 96)

6.1 Introduction to LTI Systems

Why This Matters

Every time you hear an echo in a room, adjust the equalizer on your music, apply a filter to a photo, or use noise cancellation headphones, **convolution** is happening behind the scenes. Understanding convolution allows you to analyze, design, and predict the behavior of these systems. This is one of the most important concepts in signal processing!

Linear Time-Invariant (LTI) systems are fundamental in signal and system analysis. Two key properties make LTI systems particularly important:

- Many physical processes can be modeled as LTI systems
- LTI systems can be analyzed in considerable detail, providing powerful tools

Examples of physical processes modeled as LTI:

- RC/RL/RLC circuits around an operating point
- Mass–spring–damper mechanical systems
- Audio reverb and echo effects
- Communication channel models

The Big Idea: If we know how an LTI system responds to a single impulse, we can determine how it responds to *any* input. This is the power of the **impulse response** and **convolution**.

6.2 Representation of Discrete-Time Signals in Terms of Impulses

Any discrete-time signal $x[n]$ can be expressed as a linear combination of shifted unit impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

This representation is crucial because:

- For any value of n, only one term on the right-hand side is nonzero
- The term $x[k]\delta[n-k]$ equals $x[k]$ when $n = k$ and zero otherwise

6.2.1 Reminder: The Discrete Unit Impulse Function

The discrete unit impulse (unit sample) function is defined as:

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$$

Key properties:

- **Sifting property** (extracts a value from a sum):

$$\sum_{n=-\infty}^{\infty} x[n]\delta[n-k] = x[k]$$

  Intuition: The impulse "sifts out" only the value of $x[n]$ where $n = k$.

- **Shifted impulse:** $\delta[n-k]$ is 1 at $n = k$ and 0 otherwise

- **Identity for convolution:**

$$x[n] * \delta[n] = x[n]$$

  Intuition: Convolving with an impulse returns the original signal (like multiplying by 1).

- **Sampling property** (multiplication, not summation):

$$x[n]\delta[n] = x[0]\delta[n]$$

  Intuition: When you multiply $x[n]$ by $\delta[n]$, the result is nonzero only at $n = 0$, where it equals $x[0]$.

  More generally: $x[n]\delta[n-k] = x[k]\delta[n-k]$

Common Mistake

Don't confuse the **sifting property** (involves a sum) with the **sampling property** (involves multiplication). They look similar but are different operations!

6.2.2 Graphical Interpretation

Consider an arbitrary DT signal $x[n]$. We can decompose it into a sum of weighted, shifted impulses:

$$x[n] = \cdots + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + x[2]\delta[n-2] + \cdots$$

Each term $x[k]\delta[n-k]$ represents the value of the signal at time k, shifted to position $n = k$.

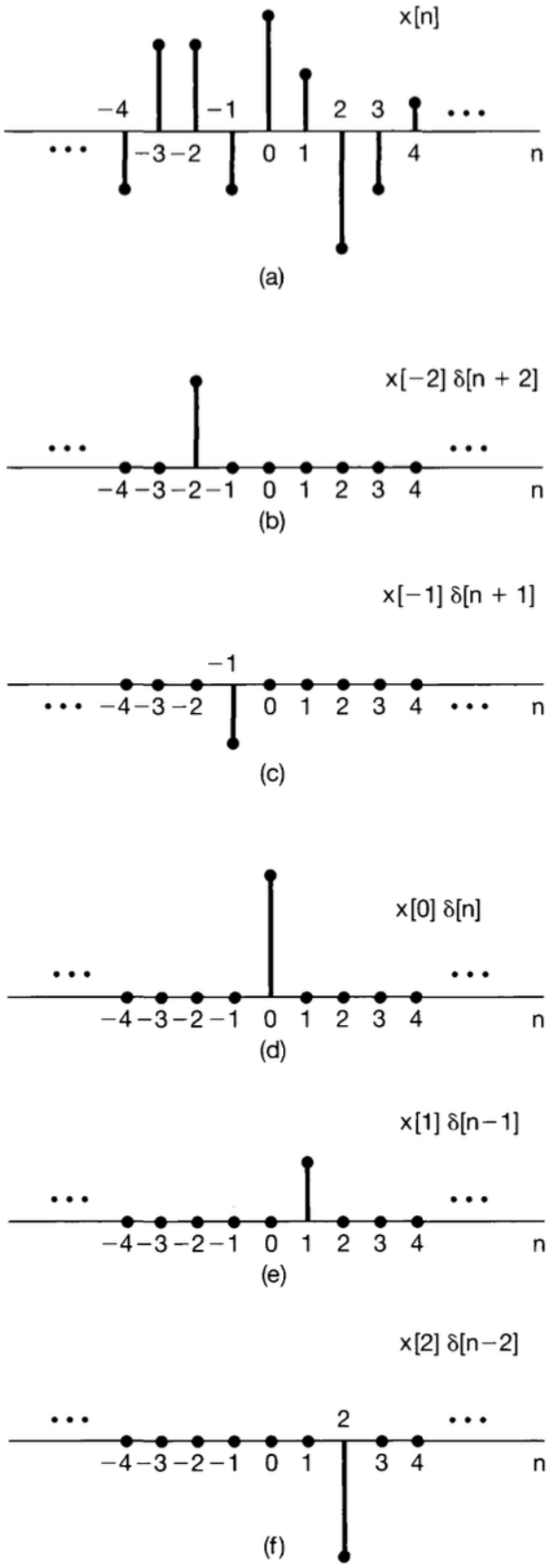

Figure 2.1 Decomposition of a discrete-time signal into a weighted sum of shifted impulses.

Figure 1: Figure 2.1: Decomposition of a discrete-time signal into a weighted sum of shifted impulses.

6.3 The Discrete-Time Unit Impulse Response

For an LTI system, if we know its response to a unit impulse $\delta[n]$, we can determine its response to any input.

6.3.1 Definition of Impulse Response

The **unit impulse response** $h[n]$ of a discrete-time LTI system is defined as the output of the system when the input is $\delta[n]$:

$$h[n] = T\{\delta[n]\}$$

where $T\{\cdot\}$ represents the system transformation.

6.3.2 Response to Shifted Impulses

Due to time invariance, if the input is $\delta[n-k]$ (a shifted impulse), the output is $h[n-k]$ (a shifted impulse response):

$$T\{\delta[n-k]\} = h[n-k]$$

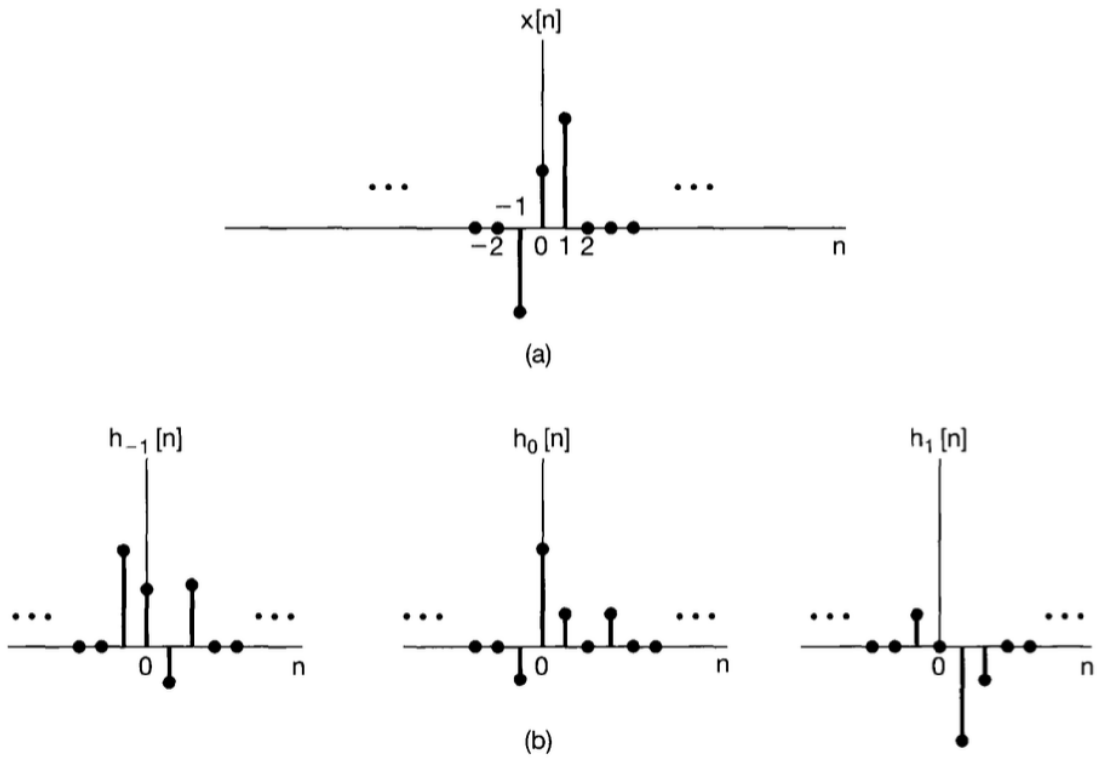

Figure 2.2 Graphical interpretation of the response of a discrete-time linear system as expressed in eq. (2.3).

Figure 2: Figure 2.2: Graphical interpretation of the response of a discrete-time linear system (part 1).

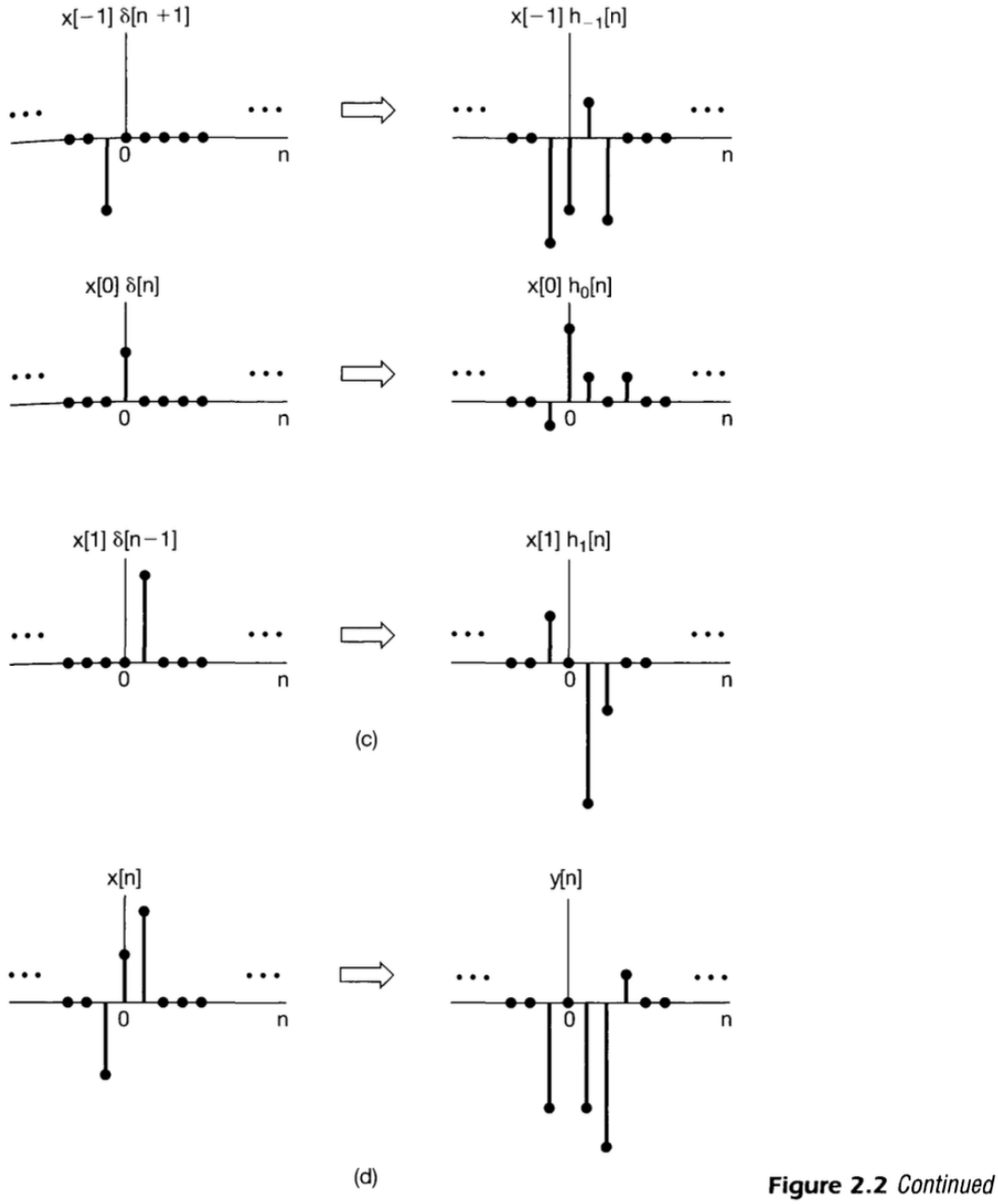

Figure 3: Figure 2.2 (continued): Graphical interpretation showing how individual responses sum to form the total output.

6.4 The Convolution Sum

6.4.1 Derivation of the Convolution Sum (Step by Step)

This derivation is important—it shows *why* convolution works for LTI systems.

Step 1: Decompose the input into impulses

Any input $x[n]$ can be written as a weighted sum of shifted impulses:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

Step 2: Apply the system to this decomposed input
The output is:

$$y[n] = T\{x[n]\} = T\left\{\sum_{k=-\infty}^{\infty} x[k]\delta[n-k]\right\}$$

Step 3: Use LINEARITY
Because the system is linear, we can move the sum and the constants $x[k]$ outside:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] \cdot T\{\delta[n-k]\}$$

Step 4: Use TIME-INVARIANCE
Because the system is time-invariant, the response to a shifted impulse $\delta[n-k]$ is a shifted impulse response $h[n-k]$:

$$T\{\delta[n-k]\} = h[n-k]$$

Step 5: Combine to get the convolution sum

$$\boxed{y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]}$$

This is called the **convolution sum** and is denoted as:

$$y[n] = x[n] * h[n]$$

Key Insight

The convolution sum tells us: **The output of an LTI system is completely determined by its impulse response $h[n]$.** If you know $h[n]$, you can find the output for any input!

6.4.2 Two Ways to Interpret Convolution

Interpretation 1: Sum of Echoes (Input-Side View)
Think of each input sample $x[k]$ as triggering its own "echo" of the impulse response. The output is the sum of all these echoes:

- Input $x[0]$ triggers response $x[0] \cdot h[n]$ (starting at $n = 0$)
- Input $x[1]$ triggers response $x[1] \cdot h[n-1]$ (starting at $n = 1$)
- Input $x[2]$ triggers response $x[2] \cdot h[n-2]$ (starting at $n = 2$)

- ... and so on

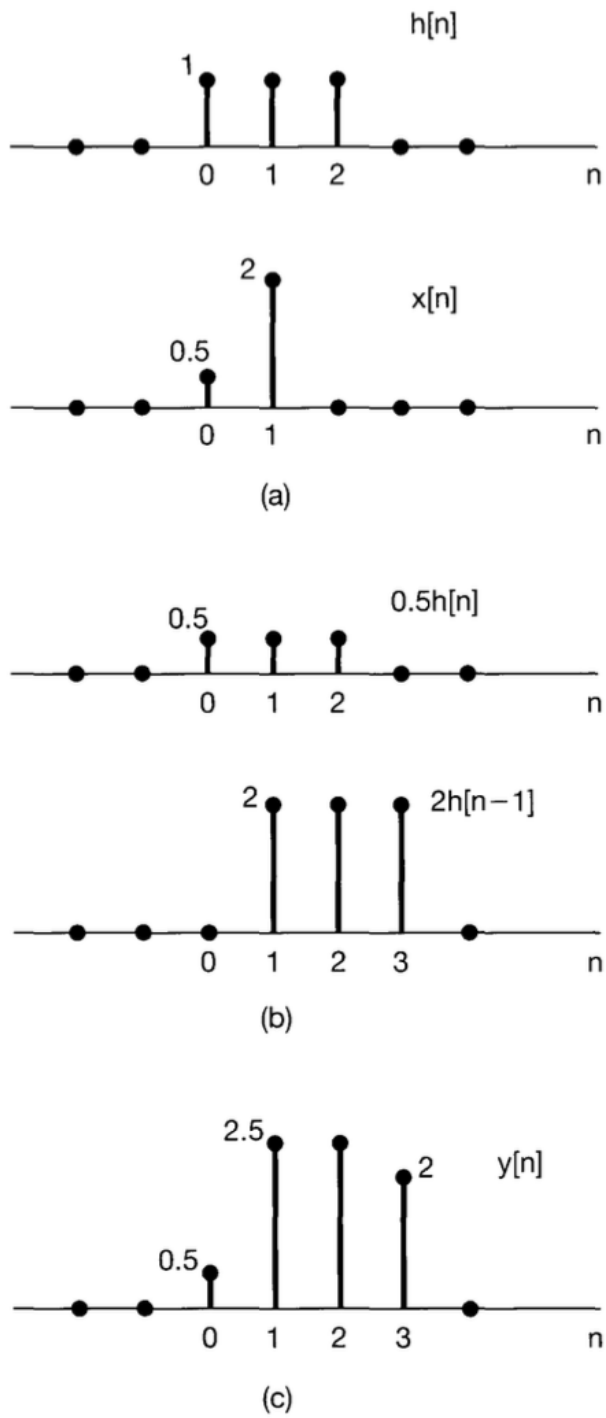

Figure 2.3 (a) The impulse response $h[n]$ of an LTI system and an input $x[n]$ to the system; (b) the responses or "echoes," $0.5h[n]$ and $2h[n-1]$, to the nonzero values of the input, namely, $x[0] = 0.5$ and $x[1] = 2$; (c) the overall response $y[n]$, which is the sum of the echos in (b).

Figure 4: Figure 2.3: The impulse response $h[n]$, input signal $x[n]$, and the individual "echoes" (responses to each impulse component) that sum to form the output $y[n]$.

Interpretation 2: Flip-and-Slide (Computational View)
To compute $y[n]$ for a specific value of n:

- **Flip**: Take $h[k]$ and flip it to get $h[-k]$
- **Shift**: Shift by n to get $h[n-k]$
- **Multiply and Sum**: Multiply $x[k]$ by $h[n-k]$ for each k and add up all products

6.4.3 Step-by-Step Convolution Procedure

To compute $y[n] = x[n] * h[n]$:

1. **Change variable**: Express both signals as functions of the dummy variable k: $x[k]$ and $h[k]$
2. **Flip one signal**: Flip $h[k]$ about $k = 0$ to obtain $h[-k]$
3. **Shift the flipped signal**: Shift $h[-k]$ by n units to get $h[n-k]$
   - If $n > 0$: shift to the right
   - If $n < 0$: shift to the left
4. **Multiply point-by-point**: For each value of k, compute the product $x[k] \cdot h[n-k]$
5. **Sum all products**: Add up all the products to get $y[n]$
6. **Repeat**: Do steps 3–5 for each value of n you need

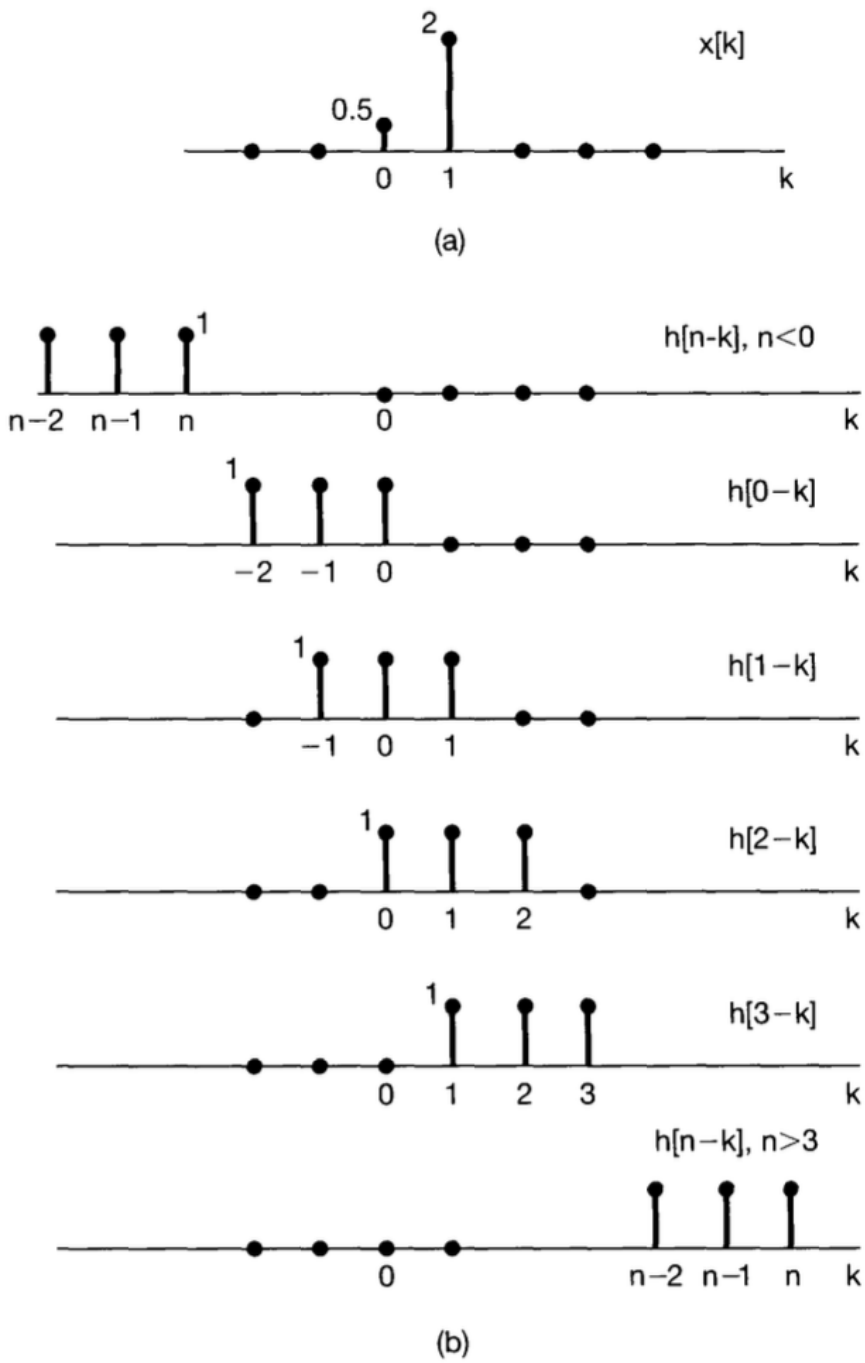

Figure 2.4 Interpretation of eq. (2.6) for the signals $h[n]$ and $x[n]$ in Figure 2.3; (a) the signal $x[k]$ and (b) the signal $h[n-k]$ (as a function of k with n fixed) for several values of n ($n < 0$; $n = 0, 1, 2, 3$; $n > 3$). Each of these signals is obtained by reflection and shifting of the unit impulse response $h[k]$. The response $y[n]$ for each value of n is obtained by multiplying the signals $x[k]$ and $h[n-k]$ in (a) and (b) and then summing the products over all values of k. The calculation for this example is carried out in detail in Example 2.2.

Figure 5: Figure 2.4: Graphical interpretation showing $x[k]$ and $h[n-k]$ for different values of n, illustrating the flip-and-shift process.

Common Mistake

A very common mistake is forgetting to **flip** before shifting. If you just shift without flipping, you'll get the wrong answer! Remember: the argument is $h[\mathbf{n} - \mathbf{k}]$, not $h[n + k]$.

6.5 Examples of Convolution

6.5.1 Worked Example: Computing Convolution Step by Step

Let's compute $y[n] = x[n] * h[n]$ for:

$$x[n] = \{1, 2, 3\} \text{ for } n = 0, 1, 2 \quad \text{(zero elsewhere)}$$
$$h[n] = \{1, 1\} \text{ for } n = 0, 1 \quad \text{(zero elsewhere)}$$

Step 1: Determine the range of the output

Key Insight

Length of convolution result: If $x[n]$ has N samples and $h[n]$ has M samples, the output $y[n]$ has $N + M - 1$ samples.

Here: $N = 3$, $M = 2$, so $y[n]$ has $3 + 2 - 1 = 4$ samples.

The output ranges from $n = 0$ (first nonzero of x plus first nonzero of h) to $n = 3$ (last nonzero of x plus last nonzero of h: $2 + 1 = 3$).

Step 2: Compute each output sample

For each value of n, we compute $y[n] = \sum_k x[k]h[n - k]$.

For $n = 0$:

$$\begin{aligned} y[0] &= \sum_k x[k]h[0 - k] = \sum_k x[k]h[-k] \\ &= x[0]h[0] + x[1]h[-1] + x[2]h[-2] \\ &= (1)(1) + (2)(0) + (3)(0) = \mathbf{1} \end{aligned}$$

(Only $h[0] = 1$ is nonzero; $h[-1]$ and $h[-2]$ are outside the support of h)

For $n = 1$:

$$\begin{aligned} y[1] &= \sum_k x[k]h[1 - k] \\ &= x[0]h[1] + x[1]h[0] + x[2]h[-1] \\ &= (1)(1) + (2)(1) + (3)(0) = 1 + 2 = \mathbf{3} \end{aligned}$$

For $n = 2$:

$$\begin{aligned} y[2] &= \sum_k x[k]h[2 - k] \\ &= x[0]h[2] + x[1]h[1] + x[2]h[0] \\ &= (1)(0) + (2)(1) + (3)(1) = 2 + 3 = \mathbf{5} \end{aligned}$$

For $n = 3$:

$$\begin{aligned} y[3] &= \sum_k x[k]h[3-k] \\ &= x[0]h[3] + x[1]h[2] + x[2]h[1] \\ &= (1)(0) + (2)(0) + (3)(1) = \mathbf{3} \end{aligned}$$

Result:

$$\boxed{y[n] = \{1, 3, 5, 3\} \text{ for } n = 0, 1, 2, 3}$$

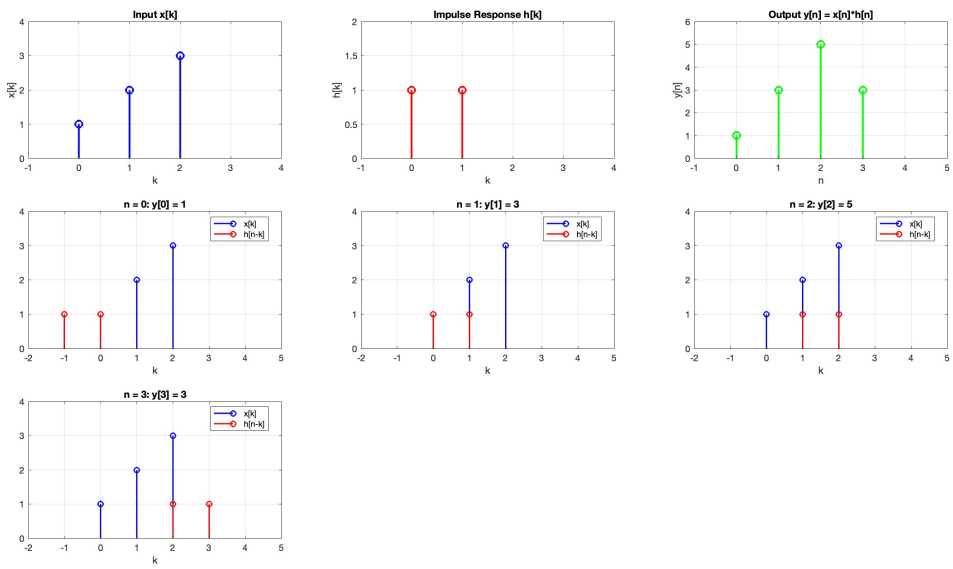

Figure 6: Graphical computation of $y[n] = x[n] * h[n]$ for $x[n] = \{1, 2, 3\}$ and $h[n] = \{1, 1\}$. Top row shows the input, impulse response, and output. Bottom panels show the flip-and-slide process for each output sample.

Key Insight

Quick check: The sum of all input samples times the sum of all impulse response samples should equal the sum of all output samples:

$$(1 + 2 + 3) \times (1 + 1) = 6 \times 2 = 12 \quad \checkmark$$

$$1 + 3 + 5 + 3 = 12 \quad \checkmark$$

This is a useful sanity check for your calculations!

6.5.2 Example: Graphical Convolution

Consider $x[n]$ and $h[n]$ with finite support. The convolution sum becomes a finite sum over the range where both signals are nonzero.

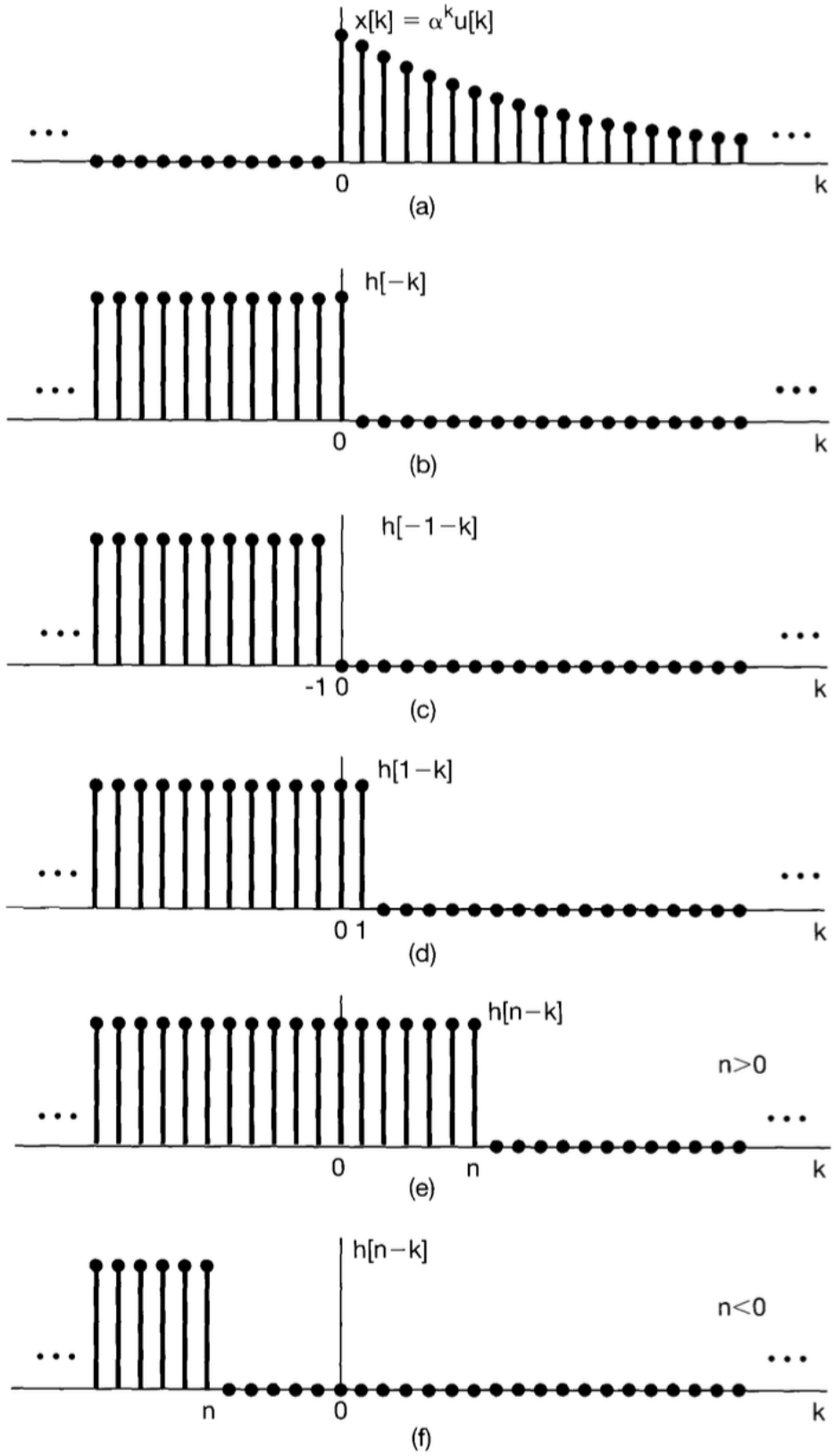

Figure 2.6 Graphical interpretation of the calculation of the convolution sum for Example 2.3.

Figure 7: Figure 2.6: Graphical interpretation of the calculation of the convolution sum, showing the flip-and-shift operation for different values of n.

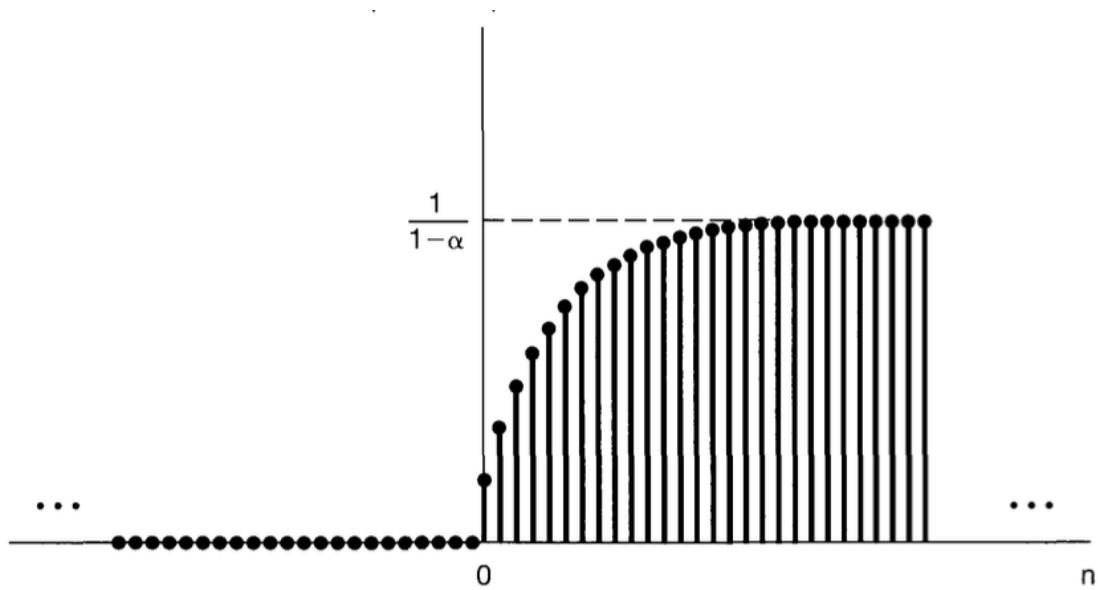

Figure 2.7 Output for Example 2.3.

Figure 8: Figure 2.7: Output for Example 2.3, showing the result of the convolution calculation.

6.5.3 Example: Convolution with the Unit Step (Accumulator)

If $h[n] = u[n]$ (unit step), then the convolution gives:

$$y[n] = x[n] * u[n] = \sum_{k=-\infty}^{n} x[k]$$

This represents the **accumulation** (running sum) of the input signal.
Why does this happen? Since $u[n-k] = 1$ for all $k \leq n$ and $u[n-k] = 0$ for $k > n$:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]u[n-k] = \sum_{k=-\infty}^{n} x[k] \cdot 1 = \sum_{k=-\infty}^{n} x[k]$$

Example: If $x[n] = \{1, 2, 3\}$ for $n = 0, 1, 2$:

$$\begin{aligned}
y[0] &= x[0] = 1 \\
y[1] &= x[0] + x[1] = 1 + 2 = 3 \\
y[2] &= x[0] + x[1] + x[2] = 1 + 2 + 3 = 6 \\
y[3] &= x[0] + x[1] + x[2] + x[3] = 6 + 0 = 6 \quad \text{(stays at 6)}
\end{aligned}$$

6.5.4 Example: Convolution with a Shifted Impulse (Delay)

If $h[n] = \delta[n - n_0]$ (a shifted impulse), then:

$$y[n] = x[n] * \delta[n - n_0] = x[n - n_0]$$

The output is simply the input **delayed** by n_0 samples. This is why $\delta[n - n_0]$ is called a "delay element."

6.6 Properties of the Convolution Sum

These properties are extremely useful for simplifying calculations and understanding system interconnections.

6.6.1 Commutativity

$$x[n] * h[n] = h[n] * x[n]$$

Meaning: You can flip either signal when computing convolution—the result is the same. This is useful because you can choose to flip whichever signal is simpler.

6.6.2 Associativity

$$(x[n] * h_1[n]) * h_2[n] = x[n] * (h_1[n] * h_2[n])$$

Meaning: For cascade (series) connections of LTI systems, the order doesn't matter, and you can combine the impulse responses first:

$$h_{\text{equivalent}}[n] = h_1[n] * h_2[n]$$

6.6.3 Distributivity

$$x[n] * (h_1[n] + h_2[n]) = x[n] * h_1[n] + x[n] * h_2[n]$$

Meaning: For parallel connections of LTI systems, you can add the impulse responses:

$$h_{\text{equivalent}}[n] = h_1[n] + h_2[n]$$

6.6.4 Identity Element

$$x[n] * \delta[n] = x[n]$$

Meaning: Convolving with $\delta[n]$ leaves the signal unchanged (like multiplying by 1).

6.6.5 Shift Property

$$x[n] * \delta[n - n_0] = x[n - n_0]$$

Meaning: Convolving with a shifted impulse delays the signal by n_0 samples.

6.7 Summary and Key Formulas

Concept	Formula/Definition
Signal decomposition	$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
Impulse response	$h[n] = T\{\delta[n]\}$
Convolution sum	$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$
Output length	If $x[n]$ has N samples, $h[n]$ has M samples, then $y[n]$ has $N+M-1$ samples
Commutativity	$x[n] * h[n] = h[n] * x[n]$
Associativity	$(x * h_1) * h_2 = x * (h_1 * h_2)$
Distributivity	$x * (h_1 + h_2) = x * h_1 + x * h_2$
Identity	$x[n] * \delta[n] = x[n]$

6.8 Common Mistakes to Avoid

1. **Forgetting to flip:** The most common error! Remember: $h[n-k]$ means flip $h[k]$ first, then shift.

2. **Wrong limits:** When computing the sum, only include terms where *both* $x[k]$ and $h[n-k]$ are nonzero.

3. **Confusing n and k:**

   - n is the output time index (the variable you're solving for)
   - k is the dummy summation variable

4. **Forgetting signal supports:** Always identify where each signal is nonzero before computing.

5. **Sign errors with shifts:** $h[n-k]$ for $n=3$ and $k=1$ gives $h[3-1] = h[2]$, not $h[-2]$ or $h[4]$.

Key Insight

Sanity check: Always verify that $\sum_n y[n] = (\sum_n x[n]) \cdot (\sum_n h[n])$. If this doesn't hold, you made an error somewhere!

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